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Quantum-inspired Computing: Entanglement-enhanced Technique for Short Portfolio in Global Markets

Yu-Chi Jiang, Yun-Ting Lai, Po-Chun Chen, Yu-Yu Chang, Kun-Min Wu, Shu-Yu Kuo, Yao-Hsin Chou, and Sy-Yen Kuo

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***Abstract*—Portfolio optimization is an essential aspect of the development of quantum computing. The quantum-inspired optimization (QIO) algorithm facilitates an efficient search for the optimal solution by simulating quantum mechanics on a classical computer, thus bridging quantum theory and the actual implementation of quantum computing. This study proposes an entanglement-based QIO to optimize the short-selling portfolio in a group of seven (G7) industrialized nations, which are the world's largest markets and significantly impact global economies. To diversify investment options in response to the ever-changing markets, short-selling is a worthy topic for discussion. The innovative trend ratio can precisely determine the performance of a short-selling portfolio during a stable downward trend. Implementing the short-selling trend ratio model in the significant G7 markets broadens its applicability.**

I. Introduction

Quantum computing is an essential developed field for demonstrating quantum technology's superiority. Quantum search algorithms are among the applications [1, 2] where the quantum computer outperforms the classical computer. Real-world applications, such as portfolio optimization [3], can greatly benefit from quantum computing's parallelism advantage. Nevertheless, the current quantum computer has lower fidelity, coherence time, and fault tolerance [4]. These limitations render the quantum computer noisy and prevent its generalized application. The quantum-inspired optimization (QIO) algorithm is one of the efficient quantum computing techniques that can reap the quantum benefits of optimizing complex problems by simulating quantum mechanics on a conventional computer. A QIO algorithm could bridge the technology gap between the quantum theory of quantum computation and its application to real-world, large-scale problems before the widespread use of the quantum computer.

Because of the ever-changing financial market and the vast number of stocks, portfolio optimization is a complex and computationally intensive real-world application problem. The investor attempts to profit from diversified investment strategies in the turbulent market. Long-selling is the most frequently discussed strategy, with the expectation that the market would increase in value. In addition to the long-selling strategy, there is a short-selling strategy for gaining wealth and stabilizing an investment when the market is down. This strategy involves betting that the market will decline and earning the difference. Short-selling plays an important role in stabilizing the market [5].

Portfolio evaluation is a crucial aspect of computational intelligence. The evaluation indicator determines the type of portfolio that the QIO would search for. Sharpe ratio [6] is a prevalently used indicator for simultaneously assessing a portfolio's return and risk. The return employs the excess return, while the risk premium of the Sharpe ratio adopts the variance and covariance definitions of the modern portfolio theory [7]. A flat stock with low variance (i.e., low risk in the Sharpe ratio) has a high Sharpe ratio value. Instead of a flat stock, the investor requires a stable but rising stock. There are numerous studies aimed at improving the Sharpe ratio. Fuzzy Sharpe ratio [8] addresses the market return uncertainty by the fuzzy model, and the Sortino ratio [9] considers the downside risk. In recent years, a novel trend ratio indicator has been proposed [3], indicating a significant improvement over the Sharpe ratio. The trend ratio utilizes funds standardization [10] to convert the stock price into the fund's volatility and then uses the linear regression line as the basis for return and risk evaluation. Risk is the difference between the regression line and the portfolio trend. The trend ratio enables the search for the investor-desired portfolio with a stable upward trend in studies [3, 11] and a stable downward trend in studies [12].

This study evaluates short-selling portfolios based on the trend ratio. A portfolio with a stable uptrend or downtrend can be effectively constructed using the trend ratio. The duration of the evaluation is an additional crucial setting. The sliding window can aid in determining the optimal period for the portfolio construction of short-selling strategies. The mechanism retrieves the most up-to-date data as the training set and preserves the training data's freshness without an outdated reference. Due to the global impact of COVID-19, this study generalizes the intelligent short-selling portfolio optimization system to the most influential stock market. The purpose is to demonstrate that the short-selling strategy assessed by the trend ratio and sliding window mechanism is applicable to major markets worldwide. The most influential market in the world is a member of the Group of Seven (G7), an annual forum for discussing financial and currency policies. As of October 2022, according to Statista, G7 members account for more than half of the world's market capitalization. Consequently, the G7 market is an important target in the financial market discussion. Study [13] investigates the relationship between political risk across G7 countries and the hedge fund returns because the G7 forum helps stabilize global political institutions, and their action can impact global economies. Study [14] discusses the financial contagion phenomenon between the G7 market and China during the COVID-19 period. The discussion of the G7 market is a prevalent issue.

This study applies a QIO algorithm among G7 intermarkets as an intelligent short-selling strategy system to develop a more diverse strategy. QIO is an emerging optimization strategy inspired by nature. The classical nature-inspired evolutionary search optimization methods for portfolio optimization are genetic algorithm (GA) [10, 15], particle swarm optimization (PSO) [16], and artificial bee colony optimization (ABC) [17].

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The QIO proves to be more effective in portfolio optimization than traditional GA in study [3]. One of the effective QIO is the quantum-inspired tabu search algorithm (QTS) [18]. There are many complicated optimization problems addressed by QTS, including classical knapsack problem [18], traveling salesman problem [19], and quantum [20] and boolean reversible circuit synthesis [21, 22]. The financial problems are a more challenging issue, and thus the global best solution [23] and the quantum-not gate mechanism [24] is proposed to enhance the effectiveness and efficiency of QTS. The first contribution of this paper is the first to apply the QIO algorithm and the innovative trend ratio to representative G7 countries in the short-selling strategy. Second, this study proposes a novel entanglement local search-assisted (ELSA) mechanism to improve QTS for constructing the optimal short-selling portfolio based on a novel evaluation of trend ratios.

II. Proposed Method

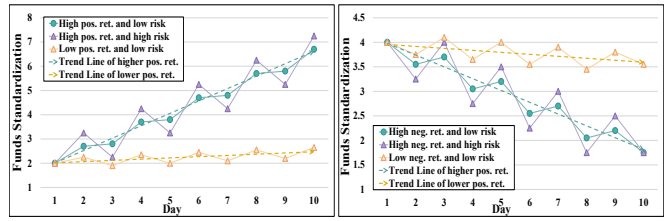
A. Trend Ratio [3] with Short-selling

Trend Ratio [3] (TR) is a novel assessment indicator that can evaluate the return and risk of the portfolio through the trend line. The slope of the trend line is the daily expected return, and the volume of daily funds standardization [10] deviation from the trend line is a risk. The purpose of the TR is to maximize the daily return when taking a unit of risk. In Fig. 1(a), if two portfolios have the same risk, the copper-green one with a higher expected return has a steeper trend line slope and a higher TR than the yellow one. Conversely, if two portfolios have the same expected return, the copper-green one with a lower deviation from the trend line has lower risk and higher TR than the purple one.

The stock market fluctuates and is influenced by the behavior of investors. When the stock market is in an uptrend, it is suitable to be involved in normal trading (long selling). In contrast, short-selling is appropriate when the stock market is in a downtrend. When investors predict that a stock will decline, they borrow it from the stock vendor at a relatively high price and then sell it in advance. When the stock goes down and hits a relatively low point, investors will buy the stock back and return it to the lender. Investors earn a profit from the price difference. Short-selling does not mean investors should select a portfolio with a poor performance that has a high risk or low return. The TR also can assess the stable downtrend portfolio, which is the opposite trend of normal trading; having lower positive risk and more negative return is better, as shown in Fig.1(b). In the short-selling trend ratio evaluation, a steeper slope of the trend line corresponds to a greater return, i.e., a more negative return. Even in a short-selling strategy, it is better for a stable trend close to the trend line. The copper-green short-selling strategy can achieve a higher trend ratio with a stable downward trend and a steeper slope of the trend line portfolio.

B. Entanglement Local Search-Assisted Technique Based on QIO Algorithm

Finding the optimal portfolio for each G7 nation is a difficult combinatorial problem. For robust searching efficiency, this



(a) Long selling (b) Short selling

Fig. 1: In both long and short selling, portfolios guided by trend ratio have steeper trend lines and less volatility simultaneously, which best satisfies investors' expectations.

study proposes quantum-inspired computing with the entanglement local search-assisted technique (ELSA) to promote search effectiveness. The adopted QIO is based on the global-best guided quantum-inspired tabu search with a quantum-Not Gate (GNQTS). First, ELSA-GNQTS inherits the characteristics of QTS, which approaches the best solution and remains far from the worst ones. Nevertheless, with global best solution guidance and Quantum-Not gate techniques based on GNQTS, ELSA-GNQTS also has outstanding abilities in both exploitation and exploration. In addition, to effectively refine solution quality, ELSA-GNQTS employs an entanglement local search-assisted method to search for potential neighbor portfolios that have entanglement-like relationships with the local optima. ELSA-GNQTS details are below:

1) Problem encoding: ELSA-GNQTS uses binary encoding solution S, which represents the selection states of stocks as either 0 (excluded) or 1 (included). The length of each S corresponds to the number of stocks used in each G7 market, represented by n.

2) Initialization: ELSA-GNQTS uses beta-matrix b to represent probabilities of stocks being selected, as it is inspired by the quantum superposition state shown in Eq. (**??**). All probabilities b_i in beta-matrix are set to be 0.5 initially, which means stocks have both 50% chances to be added into the portfolio or not respectively. The probability of stock i represents by $b_i = \langle\psi_i|1\rangle\langle 1|\psi_i\rangle = \beta_i^2$.

3) Measurement: Measurement makes the quantum state collapse into a particular state. ELSA-GNQTS compares the probability b_i in beta-matrix with randomly generated floating numbers. The ith stock will be the component of the portfolio if b_i is bigger than random numbers. Eq. (1) illustrates how the measurement process executes. The portfolio is determined by the measured state, $P = \{stock(S_i)|S_i = 1, i \in [1, n]\}$. The fitness of a solution (portfolio) is the trend ratio.

$$S_i = \begin{cases} 1, & \text{When } b_i \geq Random\ Number \\ 0, & \text{When } b_i < Random\ Number \end{cases} \tag{1}$$

4) Quantum Not Gate: The indispensable mechanism making ELSA-GNQTS have a better explore ability to escape local optima toward global optima is the Quantum Not (QN) gate. The QN gate applies as $\alpha|0\rangle + \beta|1\rangle \xrightarrow{QNgate} \beta|0\rangle + \alpha|1\rangle$. This procedure swaps the probabilities of a state is 0 and 1. According to the selection state of the global best (S^{Gb}) and the local worst portfolio (S^{Lw}), when it comes to discrepant decisions about stock i, e.g., $S_i^{Gb} = 1$ and $S_i^{Lw} = 0$, and the probability favors the worst outcome, e.g., $b_i < 0.5$, the QN gate will be activated and reverse probability b_i. All

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circumstances are shown in Eq. (2). When all b_i are examined, the beta-matrix makes the measurement results at succeeding iterations more compatible with the optimal portfolio.

$$if \begin{cases} S_i^{Gb} = 1 \ \& \ S_i^{Lw} = 0 \ \& \ b_i < 0.5 \\ S_i^{Gb} = 0 \ \& \ S_i^{Lw} = 1 \ \& \ b_i > 0.5 \end{cases}, \ b_i = 1 - b_i \tag{2}$$

5) Updating: This step adjusts the beta-matrix's probabilities b_i according to the components that the global best and the local worst portfolios selected in Eq. (3). When selection is different, the probability of the global best portfolio will increase, and the local worst one will reduce inversely.

$$if \begin{cases} S_i^{Gb} = 1 \ \& \ S_i^{LW} = 0 \ , \ b_i = b_i + \theta \\ S_i^{Gb} = 0 \ \& \ S_i^{LW} = 1 \ , \ b_i = b_i - \theta \end{cases} \tag{3}$$

6) Entanglement Local Search-Assisted (ELSA) Mechanism: Local search is a technique that improves the quality of a solution by examining the neighborhood area surrounding a local optimum. The neighborhood means the closest solution. In portfolio optimization, this study finds that stocks have a dependency relation. Certain industries, for instance, have varying degrees of cooperation and competition. The optimal portfolio constructions reflect these relationships. For instance, stocks that act as complements are frequently present or absent from portfolio components simultaneously, whereas two stocks with a similar trend would choose either. This study refers to these stock relationships in portfolio optimization as neighborhood selection. The neighborhood relation in quantum-inspired computing can be simulated as the entanglement relation. This study proposes the ELSA mechanism to search for a better short-selling solution and the dependency relation.

The considered neighborhood area includes two categories, where S^{nb} represents the neighborhood solution: (a) the basic neighbor of the binary encoding, the Hamming distance is 1 ($\sum_{i=1}^{n}(S_i^{Gb} \oplus S_i^{nb}) = 1$), i.e., local optimum adds a stock (+1) or removes a present stock (-1), (b) stocks with the entanglement relation, including $|00\rangle + |11\rangle$, $|01\rangle + |10\rangle$, $|001\rangle + |110\rangle$, and $|0011\rangle + |1100\rangle$. ELSA implementation searching for neighborhood solutions is to add two stocks (+2) or remove two stocks (-2), add a stock and remove an existing stock (+1-1), add two stocks and remove an existing stock (+2-1), add a stock and remove the existing two stocks (+1-2), and add two stocks and simultaneously remove two existing stocks (+2-2). These neighbors satisfy Eq. (4) , where the $n(P)$ represents the number of stocks in the portfolio S. Numerous real-world applications have dependency relationships; therefore, when the domain dependency information can be clearly observed and stated, the entanglement relation can improve solution quality via the ELSA technique. After termination criteria are satisfied, ELSA-GNQTS searches for entangled neighbors close to the current best solution. ELSA-GNQTS keeps exploiting potential optima around the current global best until there are no better neighbor portfolios around the latest best solution.

$$\begin{matrix}\text{Entanglement} \\ \text{Neighborhood}\end{matrix} \in \begin{cases} \sum_{i=1}^{n}(S_i^{Gb} \oplus S_i^{nb}) = 2 \\ \sum_{i=1}^{n}(S_i^{Gb} \oplus S_i^{nb}) = 3 \text{ and } |n(P^{Gb}) - n(P^{nb})| = 1 \\ \sum_{i=1}^{n}(S_i^{Gb} \oplus S_i^{nb}) = 4 \text{ and } n(P^{Gb}) = n(P^{nb}) \end{cases} \tag{4}$$

C. Sliding Window

In this study, we utilize the TR and the proposed intelligent system, ELSA-GNQTS, to assess the portfolio in short-selling strategy in the stock market, aiming to find a stable downtrend portfolio with low risk and high return. However, there is a great deal of historical information about the stock price in the stock market. Adopting years of data as a training period for a day or a week is unreasonable and vice versa.

This paper utilizes a sliding window to select an appropriate training-to-testing period. We adopt common financial reporting periods as the rebalancing frequency: year (Y), half year (H), a quarter (Q), and month (M). Regarding the year-on-year category, some industries might experience phenomena related to their business cycle. Therefore, utilizing the training data from the same period of the previous year is favorable. With the assistance of these sliding windows, we can maintain freshness and eliminate outdated training data.

III. Experimental Result

A. Experimental Environment

The proposed intelligent short-selling system is implemented in the G7 stock markets using the trend ratio model and the ELSA-GNQTS. The G7 is an important target to prove the generality of the trend ratio usage in the short-selling strategy in the stock market. The investment period for this study comprises from January 2013 to December 2022, encompassing the before, during, and after the COVID-19 pandemic to demonstrate the generality. To fairly compare the performance of these countries, this study selects the thirty largest capitalization stocks with complete ten-year price data from these indexes. TradingView and Investing.com provide the capitalization data. The stock price data comes from Wharton Research Data Service (WRDS), Yahoo Finance, and Investing.com. The parameters of the conducted ELSA-GNQTS are as follows: 10 individuals, 10,000 generations, 50 independent experiments, and 1 billion local currency initial funds. The theta setting is taken from [25], as 0.0002. The final result takes the best solution from 50 independent experiments.

B. Portfolio Performance Analysis

First, this section compares the performance of risk and trend ratio between portfolios and single stocks. A return evaluation for the short-selling strategy converts to a positive trend ratio; therefore, the higher the trend ratio, the better. The definition of a single stock is the most stable downtrend in each training-to-testing period, that is, the individual stock having the highest trend ratio. The portfolio selected by ELSA-GNQTS is the combination with the highest trend ratio in each period. The experimental results for each market are the average of the total training-to-testing periods. Fig. 2(a) shows the risk performance of the portfolio and the single stock across seven countries. The portfolio risk is lower than the single stock risk. The result proves that the portfolio can spread the risk more effectively than a single stock in all G7 countries. FTSE achieves the lowest portfolio risk, and the DJIA gets the lowest single stock risk.

Regarding trend ratio performance in Fig. 2(b), the portfolio achieves a higher trend ratio in seven markets. The FTSE achieves the highest trend ratio and lowest risk of short-selling portfolios among the G7 countries, which means that the UK is suitable for diversified investment. A positive correlation exists

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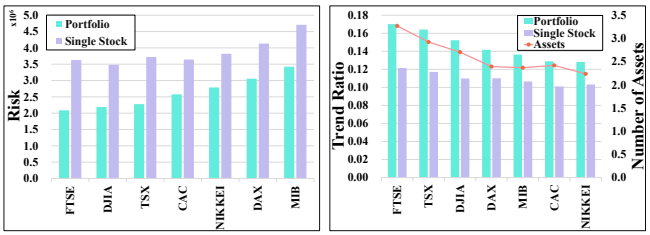
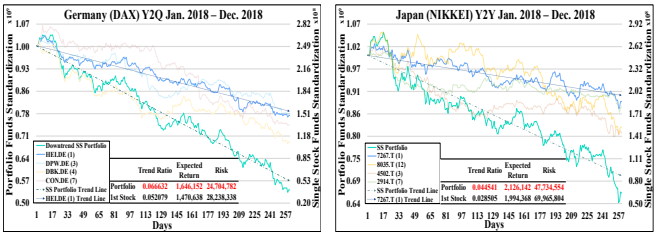


(a) Risk comparison (b) Trend ratio comparison

Fig. 2: Comparisons between portfolio and single stock performance among G7 countries (a) The portfolio carries less risk than a single stock. The FTSE achieves the top rank regarding portfolio risk, and DJIA has the lowest risk in the single stock. (b) The portfolio trend ratio is higher than that of a single stock. The number of assets and the trend ratio are strongly related.



(a) DAX (b) NIKKEI

Fig. 3: The best stable downtrend portfolios (copper-green) include the top-ranked stock in general. The portfolios have a superior trend ratio, expected return, and risk performance than the top-ranked stock in the illustration.

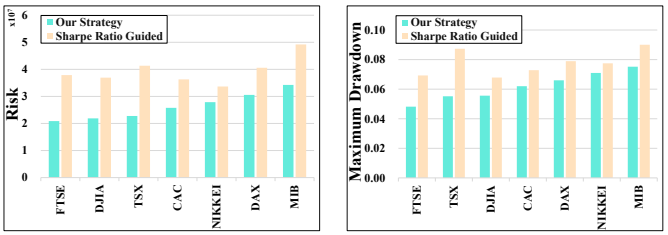
between the number of assets and the trend ratio; the optimal portfolio with a higher trend ratio requires more stocks.

A portfolio comprises several assets that can lower risk and improve trend ratio performance. In these markets, the trend ratio method effectively identified an investor-desired downtrend portfolio in a short-selling strategy. Due to space limitations, Fig. 3 illustrates the general cases with two markets, DAX and NIKKEI. The constituents and short-selling portfolio (SS portfolio) are presented in a downward trend, and the optimal portfolio contains the top-ranked stock.

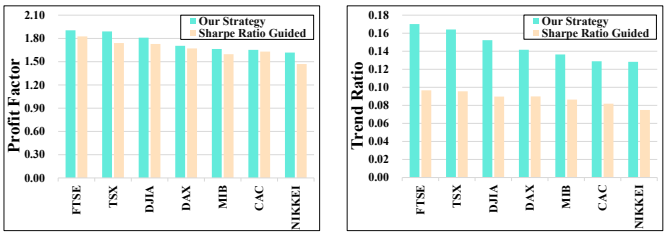
C. Benchmark Strategy Analysis

This paper establishes the intelligence model based on trend ratio and ELSA-GNQTS that composes the short-selling portfolio with high performance in G7. The Sharpe ratio is the typical indicator to evaluate portfolio performance. This section compares the portfolio performance between our strategy and the Sharpe ratio regarding the risk, trend ratio, maximum drawdown (MDD), and profit factor (PF). The MDD and PF are prevalent portfolio evaluation indicators. MDD is the risk metric used to quantify the greatest loss from peak to trough. PF is to evaluate the portfolio's profit under loss.

First, we compare the risk performance. As shown in Fig. 4(a), our strategy generates a portfolio with a lower risk than the Sharpe ratio across all G7 markets. In MDD risk evaluation aiming to assess the loss volatility, our strategy has a lower MDD across all G7 markets in Fig. 4(b), indicating that the portfolios created by our model have more stable fluctuation and lower risk. Regarding the return performance of PF evaluation in Fig. 4(c), our strategy has a higher PF in all markets than the Sharpe ratio, representing our portfolio has greater profitability. Thus, our strategy constructs the portfolio with a greater trend ratio than the Sharpe ratio in Fig. 4(d), where the trend ratio simultaneously evaluates the return and

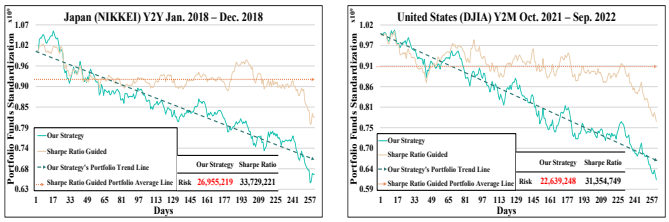


(a) Risk (b) Maximum drawdown



(c) Profit factor (d) Trend ratio

Fig. 4: Comparisons are between our strategy and the Sharpe ratio among G7 markets. Our strategy (copper-green) has an outstanding performance in terms of risk, maximum drawdown, profit factor, and trend ratio.



(a) NIKKEI (b) DJIA

Fig. 5: Portfolio guided by our strategy (copper-green) has higher return and less volatility than the Sharpe ratio. This result is consistent with Fig. 5.

risk. Besides, the FTSE of our strategy reaches the lowest risk and MDD, highest PF and trend ratio, demonstrating that FTSE has a high return and low risk in short-selling.

Fig. 5 shows the short-selling portfolios fluctuation between the trend ratio (copper-green line) and the Sharpe ratio (orange line) evaluation. According to the figure, our copper-green portfolio has a substantially strong decline tendency consistent with the design concept. As the Sharpe ratio evaluates the risk associated with the portfolio's mean value, the portfolio's trend is likely to be flat. Consequently, ELSA-GNQTS and the trend ratio form short-selling portfolios with a stable and downtrend of funds, which is consistent with the investor's psychology.

IV. Conclusion

This study presents a quantum-inspired computing ELSA-GNQTS for the global intelligent short-selling system based on the trend ratio evaluation. A steady profit short-selling portfolio can be constructed in G7 nations. The proposed ELSA technique significantly outperforms other QIO algorithms. The optimal short-selling portfolio can diversify risk better than the best single stock trade-off and achieve a higher return, lower risk, and greater trend ratio. This research expands quantum computing applications via the short-selling trend ratio model. The proposed systems in G7 countries are superior to the Sharpe ratio in terms of trend ratio, risk, MDD, and PF, demonstrating the generalizability of the short-selling strategy based on the trend ratio. Future research can be combined with quantum annealing to realize the trend ratio-based portfolio optimization system, thereby expanding the applications of quantum computing and advancing quantum technology.

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